Class-Asymmetric Robustness and Transformation-Order Effects in Lightweight Audio Deepfake Detection

Abstract

The effect of augmentations on audio deepfake classification is studied in conditions of ordered noise and Opus processing. Three pairs of MobileNetV3Small-LSTM models trained on 640 LA samples of ASVspoof 2019 reached an advantage of macro stressed ROC-AUC for paired development corpora. A nesting comparison at 9207 randomization seed increased training corpus twice along with number of optimizer iterations while keeping epochs equal to 21. An augmentation advantage of macro stressed ROC-AUC decreased from 0.1893 for 640 recordings to 0.1081 for 1,280. At the larger corpus size and threshold 0.5 the amount of stressed FAKE misses decreased from 248 to 64 of 336 views, whereas amount of stressed REAL false alarms increased from 5 to 161 of 336. The selected Channel checkpoint violated both clean-risk guards. Reversing noise and Opus changed classwise decisions on the same recordings, and the REAL error contrast varied by training protocol. The locked within-release evaluation of the original four models on 1,088 recordings showed higher stressed ROC-AUC and fewer pooled stressed FAKE misses, alongside more clean FAKE misses and more stressed misses on A17–A19.

Keywords. Forensic audio, channel enhancement, codecs, anti-spoofing measures

1 Introduction

An audio deepfake detector identifies speech for possible synthesis or voice conversion manipulation. At any point of an authentication pipeline, a false alarm sends an honest speech sample to a manual verification stage, while a missed alarm lets impersonation pass undetected. Costs vary depending on the authentication pipeline and prevalence of spoofed traffic. A model selected only according to its discrimination capacity can behave differently at a given operating point when a recording goes through the usual signal processing procedure.

Both noise and lossy compression are especially important since they could happen in any order. Recordings taken in a noisy environment can be compressed prior to transmission, and background noise could be added after decompression. Both signal processing pipelines keep the same class label of a source, but alter the data available for detection. Even under identical noise and encoder settings, the sequence of these operations alters the waveform, spectral characteristics, ranking, and threshold decisions. Measures taken between matched views help isolate the effect of signal processing order implementation for a given recording.

Six heterogeneous audio deepfake detectors are benchmarked in the accepted LAVA conference manuscript. The extension described here tackles a different problem using a matched manipulation on one architecture. For each of three initialization blocks, the set of training recordings, order, optimizer, numerical backend, and checkpointing scheme are the same. The manipulated variable is whether the training recording is seen as a clean one or one of the views from the channel bank. Both trained detectors in each initialization block are assessed for each of development recording and conditions.

Analysis will examine if the aggregated robustness is going in the same direction for all three blocks, where errors are introduced and where the attackers have to take the responsibility, along with the effect of reversal of a standard codec and noise chain on the introduced errors. Thus,

the obtained information is comparative and source-related. Sections 2-5 specify the design, and Sections 6-8 distinguish measured findings from possible explanations and threats to validity.

1.1 Operational question and scope

The subsequent action in consequence of a detector decision will determine which type of error is costly. A system that directs suspicious videos for analysis will see more videos reviewed when real audio passes through the detection threshold. A system employed to rank potential manipulations is bound to face a different loss in case of missing a synthetic or converted video. A validation cohort designed in a balanced manner will provide answer to a diagnostic question of conditional errors. In that sense, it cannot serve to measure the workload in a future deployment with unknown spoof prevalence. The quantities mentioned above are denominated by class and condition in such a way that a practitioner will be able to consider both errors.

Clean evaluation provides a valuable reference point since a detector is able to rank bona fide and spoofed files without any alteration of score distributions through channel processing. However, in practice, both types of alterations can be applied to audio production and communication in either order. It is possible to keep the source of recording, nominal noise level and codec settings unchanged while changing the order. The obtained score difference will indicate a detector sensitivity to the specified composition on the same recording. Any comparison between two different lists of recordings would mix this effect with speaker variability, content diversity, and sample differences in synthesis techniques.

Training comparison provides an answer to another question that was raised. Individual experience in exposure to different channel views can help a model when dealing with a transformed view of the evaluation set even when specific compositions were not seen during training. The experiment tests this ability in ASVspoof 2019 LA with the matched sets of training and checkpoints. It records whether the rank improvement and fixed thresholding are consistent with each other, and whether the effect remains consistent after speaker and attack-wise inspection.

1.2 Research gap and questions

The RawBoost and ADD-C approaches provide concrete means for achieving robust channels, while empirical testing confirms that genuine speech support is needed for observed mistakes [1, 2, 3]. Here are three open questions about the matching intervention and channels.

1. **Ranking and operating risk (RQ1).**

   What is the impact of switching from clean training images to sampled channel images on discrimination and false alarm rate under clean versus distorted testing? ROC-AUC analysis relies on comparing ranks across all potential operational points. Class errors relate to the decision-making process at a certain threshold. This differentiation is commonly accepted practice, but the empirical question still remains unanswered as to when do the two measures match when the training is similar.

2. **Unequal robustness (RQ2).**

   Which classes and attack IDs improve or deteriorate under the same intervention, and how consistent are the observed changes across seeds and conditions? Domain studies

Table 1: Scientific extension of the accepted conference study. Each column describes its own experimental population.

Dimension	Conference study	Journal extension
Question and models	Offline benchmark of six available detector artifacts with different training histories	Three matched 640-recording seed blocks and a nested 1,280-recording comparison at seed 9207
Cohort and stress	2,737 checksum-disjoint clean test recordings and 100 stress recordings under nine conditions	Shared development with paired noise/Opus order. Locked held-out covers 1,088 sources and four original models
Inference and uncertainty	Paired clean correctness, Holm-adjusted McNemar tests and stratified bootstrap	Class-conditional risk, attack recovery and harm, speaker-cluster intervals within each seed
Compute and scope	Historical desktop inference timing	Training logs and bounded CPU inference audit, with one architecture and within-release validation

encourage class analysis and subgroup analysis separately [3, 4]. The present source linkage distinguishes shared errors, recoveries and new errors. Speaker intervals characterize each fitted comparison. Three seed blocks and limited subgroup support constrain generalization.

3. **Processing order (RQ3).**

   How does reversing additive noise and Opus coding on the same recording affect REAL and FAKE score distributions and decisions? Noncommutativity is an established property of signal-processing compositions. Same-source pairing holds speech content, class and nominal transform settings fixed and shares the noise realization, so changes can be attributed to the tested pipeline order within each recording. Achieved distortion can differ, and this design does not identify an independent forensic cue.

1.3 Main contributions

1. **Controlled evaluation of channel augmentation.**

   Three paired MobileNetV3Small-LSTM blocks at 640 recordings share architecture, within-seed initialization, source order, optimization and the registered clean-development checkpoint rule. A nested 1,280-recording comparison measures sensitivity to the training protocol within seed 9207. The original two 640-recording pairs also have a locked within-release held-out evaluation.

2. **Class-asymmetric and attack-specific robustness.**

   We characterize false alarms and misses, including A05/A06 on development and A17-A19 on held-out, with source-level recovery and deterioration and speaker-aware uncertainty. Threshold sweeps and sub-group decomposition tests are also retrospective.

3. **Transformation-order sensitivity.**

   Pairs of Noise10-Opus24 and Opus24-Noise10 mappings find changes in the score and class-wise decisions in the same data. Matching standalone entities assist in delineating the effect of these operators.

2 Related Work

2.1 Detector architectures and benchmark design

RawNet2 learns representations straight from speech waveforms whereas AASIST learns with graph attention applied to spectro-temporal representations [5, 6]. A compact convolutional backbone represents another trade-off between compute and representations. MobileNetV3 was developed for visual recognition on mobile devices [7]. In our case, this architecture encodes consecutive Mel-based feature sequences, which are aggregated in time using a recurrent head. There exist published lightweight approaches that combine a distinct feature extraction scheme and contrastive learning [8]. The previous detector designs inform this approach, and the controlled intervention is carried out with a consistent backbone and head in all arms.

ASVspoof 2019 separates logical access, synthetic, or converted speech from physical access, replayed samples [9]. The ASVspoof 2021 campaign expands benchmarking to cover distinct logical access, physical access, and deepfake challenges under distinct conditions [10]. ASVspoof 5 scales the crowdsourced data collection and spoofs and adversary testing [11]. In-the-Wild data illustrate why a split in the benchmark dataset alone does not guarantee transport to collected in-the-wild data [12]. The current development dataset contains separated speakers from training as well as attack IDs present in training. Within the release, in terms of algorithms, A16 and A19 have commonalities with A04 and A06, respectively, and A17 with A05 [9]. Such relations place certain limitations on how held-out attack IDs can be interpreted.

The multilingual MLAAD dataset targets variations in speech synthesis systems and languages [13]. This dataset design offers an orthogonal test of transport to languages and generation domains.

2.2 Signal transformations and robust learning

Opus is a standard lossy codec [14]. Compression may affect spectral and temporal features through content-dependent coding choices. Resynthesis using a learned neural codec is another kind of transformation family, with artifacts of its own. The discussion and case study by Maltby et al. list codecs along with dataset bias and other failure modes [4]. Such work motivates a separation of operator, representation, and detector.

RawBoost mixes convolutive, impulsive, and stationary waveform distortions and demonstrates improvements on ASVspoof 2021 LA [1]. The current augmentation selects only one out of the gains, noise, codec, and clean views for each example and applies binary cross-entropy loss. Frequency-selective adversarial training modifies the selection criterion for the perturbations [15]. Group distributionally robust optimization assigns weights based on the risk of the group [16]. DK-CAST provides additional supervision by learning clean-to-distorted distillation [17]. SONAR examines spectral-contrastive residual representations across a wide range of shifts [18]. These methods tackle objective or representation changes. The current experiment keeps the representation family and the objective fixed to measure the effect of the view policy.

Table 2: Comparison of research designs. Different cohorts and operating rules prevent a direct performance ranking.

Study	Data or evaluation setting	Treatment or model	Relationship to the present comparison
RawBoost and F-SAT [1, 15]	ASVspoof and DeepFakeVox-HQ, respectively	Waveform augmentation and frequency-selective adversarial training	Establish robust training precedents. Neither is retrained as a matched arm here.
DK-CAST and SONAR [17, 18]	Speech spoofing and domain-shift evaluations	Distillation and spectral residuals	Intervene on learning objectives or representations. The present study intervenes on training views.
ADD-C and bona fide cross-testing [2, 3]	Simulated communication and diverse authentic speech	Codec and loss grids, and bona fide domain testing	Motivate stress coverage and authentic-speech reporting. The present study controls order on a fixed source.
Present study	Three development initialization blocks, shared development and locked within-release held-out	Matched lightweight detector, ordinary augmentation	Links ordered processing and class-specific error transitions on the same source. Broader transport and mechanisms remain open.

2.3 Order effects and operating-point uncertainty

ADD-C tests detection under conventional codecs and simulated packet loss and evaluates augmentation for those communication conditions [2]. Bona fide cross-testing varies authentic-speech domains against spoofed sets and shows why authentic-speech coverage matters [3]. Both motivate class- and condition-specific evaluation. Here the comparison reverses noise and conventional Opus on the same source with a shared noise plan, and measures how a matched training-view intervention changes the resulting threshold errors. This narrower intervention yields source-level evidence about the two orders under the tested settings.

Calibration concerns the interpretation of scores, while selective prediction introduces a reject action [19, 20]. Work on calibrated audio deepfake detection illustrates the importance of these distinctions in the speech domain [21]. In this case, the sigmoid prediction is represented by p, which serves as an uncalibrated score. Rank improvement along with error increment at the same threshold is feasible. The following table (Table 2) illustrates the way the closest prior research differs from this one in terms of scope.

2.4 Channel exposure and representation choices

Work with a pretrained wav2vec 2.0 front end and data augmentation provides published evidence for a different route to addressing channel and source shift [22]. The present comparison uses a compact Mel-sequence classifier with the same representation family in both arms. This limits the treatment comparison to exposure during fitting and avoids interpreting a change

of backbone, input bandwidth, or pretraining corpus as an augmentation effect. The design also avoids comparisons with a published EER obtained under a different source partition and operating protocol.

Cross-domain multilingual benchmarks such as XMAD-Bench address variation across synthesis and language domains [23]. Such benchmarks answer a different question from a source-matched transformation test. Pairing ensures that the source is held constant throughout the processing of the channel, which aids in making the contrast of the channel more interpretable. Pairing also makes sure that the outcome is conditioned on the population of the source within the chosen cohort. The cross-domain benchmark allows a larger population but creates extra differences that need to be taken into account in a score change.

Group-robust learning prioritizes a particular group criterion and usually demands the selection of groups and weighting. On the contrary, the conventional augmentation optimization minimizes the average loss over samples. It is important due to the fact that the latter objective does not have to secure the worst subgroup even when it enhances discrimination as a whole. The journal provides the comparison of the traditional policy intervention and subgroup effect measurement. No class-dependent weighting, consistency loss, and group-robust optimization are proposed.

The previous schemes also differ from each other in terms of the statistical problem posed by the operating rule. EER computes an equal error estimate of the score distribution, while a fixed threshold allows estimating separately both false alarms and misses for each processing configuration. Therefore, we retain the registered threshold and provide the analysis of the curve near it. The attack and speaker differences depend on the model fit. They supplement the benchmark comparison without making assumptions about a different source population, intervention, or operating rule.

3 Problem Formulation

Suppose x_i is recording i from the source domain, g_i its speaker, and $y_i \in\{0, 1\}$ its class, where 0 corresponds to REAL and 1 to FAKE. Then, let $p_\theta(Tx_i) \in[0, 1]$ be the transformed score by the detector. Thus, the decision rule $\widehat{y}_{i,T} = \mathbf{1}\{p_\theta(Tx_i) \geq 0.5\}$ determines the operating point. The class-conditional risks for a fixed cohort and condition are

$$R_0(T) = \frac{\sum_{i:y_i=0} \widehat{y}_{i,T}}{n_0}, \qquad R_1(T) = \frac{\sum_{i:y_i=1}(1 - \widehat{y}_{i,T})}{n_1}. \tag{1}$$

REAL false alarms and FAKE misses correspond to false positives and false negatives, respectively. The formula for computing error rates for a series of stress conditions is to sum up the numerators across all stress conditions while dividing by n_y multiplied by the number of conditions.

The area under ROC curve (ROC-AUC) calculates the rank order of FAKE and REAL with partial credit for ties. A monotonic score transformation can preserve ordering and change the number of scores crossing 0.5. Equal error rate (EER) is summarized at the empirical ROC point closest to equality between false-alarm and miss rates, using their mean at that point. Both metrics require support from both classes. The stress macro ROC-AUC is the equally weighted

mean of ROC-AUC computed separately for the three registered stress conditions.

Let $N_i(x) = x + \nu_i$ denote insertion of the registered fixed-absolute noise for source i, and let C denote encoding and decoding with Opus. Define $T_{NC} = C \circ N_i$ and $T_{CN} = N_i \circ C$. Because C uses content-dependent analysis and lossy coding, in general $C(x + \nu_i) \neq C(x) + \nu_i$. In one case the codec processes the added noise, while in the other it does not. The contrast

$$d_i = p_\theta(T_{NC} x_i) - p_\theta(T_{CN} x_i) \tag{2}$$

measures a directional score response. A decision transition records which order alone causes an error, if either does. The contrast can reflect altered input features, score placement or cues used by the detector. An independent cue measure would be needed to interpret it as forensic-evidence retention.

3.1 Ranking and conditional risk estimate different quantities

For scores S_1 and S_0 drawn from the FAKE and REAL groups of a condition, the population form of the ranking estimand is

$$\mathrm{AUC}(T) = \Pr(S_1 > S_0) + \tfrac{1}{2} \Pr(S_1 = S_0). \tag{3}$$

The finite-sample estimate averages this comparison across class pairs [24]. Each pair contributes to ranking even when both scores lie on the same side of 0.5. Conversely, a source near 0.5 may change its decision under a small perturbation that changes almost none of the pairwise comparisons. Thus AUC and threshold errors can move in different directions. The macro stress AUC gives each of the three stress conditions equal weight. Averaging scores before ranking would be making comparisons between scores in different situations and would be computing something else. If $D_i^{(m)}(T)$ is the indicator of error using method m, the contribution to the error difference conditioned on being paired is then $D_i^{(\mathrm{channel})}(T) - D_i^{(\mathrm{ERM})}(T)$. The values taken on by this indicator are $+1$ if there is a new error, -1 if the decision was recovered, and zero if there was agreement in whether there was an error. By summing these indicators, we get a count that keeps information about which sources have changed which way. This is useful because two equal counts may represent completely different sets of errors. For an attack group, the estimand is the miss rate for that particular FAKE group.

3.2 Order effects and component diagnostics

For a frozen model, let p_0, p_N, p_C, p_{NC}, p_{CN} be the respective scores for the clean, noise-only, codec-only and two composed inputs. The signed order contrast is $p_{NC} - p_{CN}$. A complementary descriptive contrast is

$$h_i = \tfrac{1}{2}(p_{NC} + p_{CN}) - p_N - p_C + p_0. \tag{4}$$

This contrast is zero if the two component changes add on the score scale and the two orders have the same score. Departures may also arise from nonlinear features or the sigmoid score mapping. Consequently the contrast is a diagnostic of output nonadditivity, and no inference

about which speech cue was preserved follows from its magnitude. The standalone matched components are available for 9107 and 9207. The registered score and decision order contrasts are available for all three development blocks.

3.3 Decision costs and attack mixtures

Suppose a deployment has spoof prevalence π and costs c_0 and c_1 for a REAL false alarm and FAKE miss, respectively. For a fixed channel and threshold, a simple expected error cost is

$$(1 - \pi)c_0R_0(T) + \pi c_1R_1(T). \tag{5}$$

The costs and frequency are not calculated here. The expression helps understand why class-balance training and equal-weight stress macro average are not decision rules. The 0.5 rule gives a reference point for matching and lets calibration and optimal threshold choice be done in a different study.

Attack mixture is important to consider when calculating the miss rate. The formula for the attack mixture for represented attack IDs a with class-conditional weights w_a is $R_1(T) = \sum_a w_a R_{1,a}(T)$. The change in weights influences the sum even when attack miss rates are constant. The mixture in development is based on chosen data sources and repeated exposure maintains attacks' identity. So the reported aggregate applies to the mentioned mixture. The attack rows allow comparison without changing mixture weights for another population.

3.4 Matching the intervention and defining the comparison

For each training seed block indexed by b, fitting policy m, and recording i under condition T, let $s_{ibm}(T)$ be the FAKE score obtained from the selected checkpoint. A between-arm order difference differs from m but leaves i, b, and T unchanged. A within-model order difference, on the other hand, differs in T but leaves i, b, and m unchanged. The former measures the combined influence of the fitting policy and then the checkpoint choice. The latter measures the detector's reaction to the input manipulation. Mistaking the one for the other is to mistake changes in the learned weights for changes in the inputs.

A threshold transition does not require a second operating point for analysis. In the case of a REAL source, going from below to above 0.5 results in a false alarm. On the other hand, for a FAKE source, the same transition corrects a miss. As such, a directional transition can affect the two types of errors in opposite ways even if the direction is the same in both cases. The empirical problem is to know the position of each relative to the fixed threshold.

4 Matched Training Methodology

4.1 Detector and objective

The audio decoding format is mono with a sampling rate of 22.05 kHz and three-second limit using frozen P0 input contract. Six segments of 0.5 seconds are converted to log-Mel power features with 128 Mel filters, 2,048 samples Hann window and 512 samples hop. P0 normalizes the power by clipping it to 10^{-10}, scales each window by its peak value, clips the dynamic range to 80

dB and resizes and replicates the tensor into a three channel 224x224 shape. MobileNetV3Small with pretrained weights from ImageNet extracts a feature representation for each segment. LSTM with 128 cells aggregates the feature vector before binary sigmoid output.

The backbone does not include the classification head for images and uses spatial averaging pooling. It makes use of shared weight across the segments. The temporal layer generates the final representation to be passed through a rectified linear unit layer of 64 units and then a dropout layer of probability 0.4 before being passed through a sigmoid output. In partial finetuning, the last 20 layers of the backbone are updatable, with the batch normalization layers kept frozen. In warm-up, the backbone is frozen throughout.

Both training phases reduce the mean binary cross-entropy of one image from each source at each visit. Inspection of the read-only exports revealed that there are 213 exported tensors containing 1,308,401 scalar values per model. This figure includes the non-trainable state but does not give an estimate of variables updated in the fine-tuning phases. If V_i is the view and $q_i = p_\theta(V_i x_i)$, then

$$\mathcal{L}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log q_i + (1 - y_i)\log(1 - q_i)]. \tag{6}$$

The ERM branch assigns V_i to the clean view. For the channel branch, sampling is done from clean, half gain, 10 dB noise, and 24 kb/s Opus views. None of the two ordered stress chains is a training view for supervised training. Randomness of training views depends on the registered seed, whereas the rest is kept the same.

Each completed 640-recording arm had 13,440 source visits. In the order clean, half gain, Noise10, and Opus24, the realized channel counts were 3,293, 3,297, 3,349, and 3,501 for seed 9107, 3,403, 3,415, 3,338, and 3,284 for seed 9207, and 3,325, 3,384, 3,320, and 3,411 for seed 9307. The clean view was always employed by ERM. The visit counts are those that correspond to the budget. There is a difference in the clean view epochs per arm, and therefore the model used for each is a prefix of the visits. The experiment is an estimate of the view policy as a bundle.

4.2 Optimization and model selection

Adam [25] is trained over 21 epochs with 840 optimizer steps at 640 samples and 1,680 steps at 1,280 samples. Each step collects gradients over 16 microbatches from a single data source. The first epoch trains the head, while the remaining 20 epochs perform partial fine-tuning with batch-normalization parameters frozen. The training phases begin from learning rates 0.001 and 0.0001, respectively, and follow the same epoch-based schedule at each sample size. The registered CPU backend performs full-shape recomputation. The large cohort has a separate registered execution plan and the same numerical kernels and model definition. Its 82-update qualification compared uninterrupted and resumed execution across the warm-up boundary at update 80. Restored weights, optimizer state, histories and predictions matched. Full development inference parity covered 2,576 scores with zero threshold flips.

Validation tier A evaluates clean mixed-class development at each epoch. Tier B also measures the registered stress and REAL-only endpoints every third epoch and at completion.

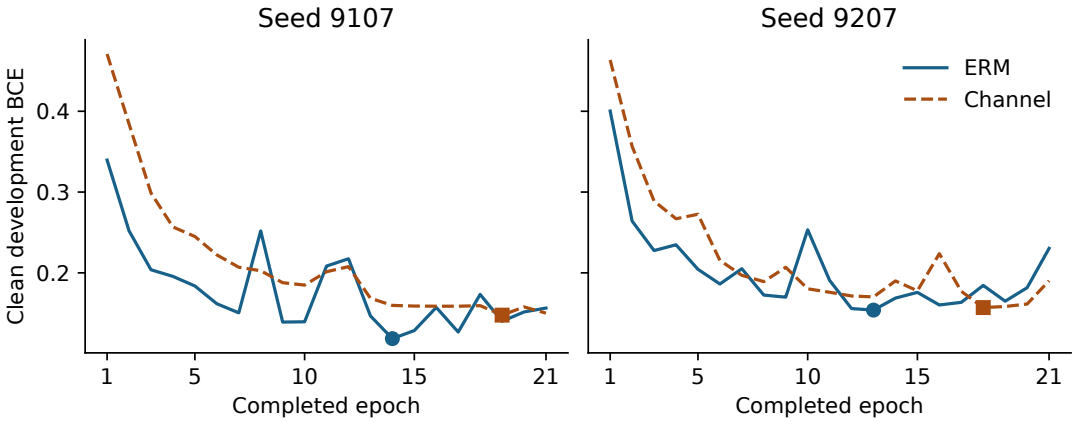


Figure 1: Recorded clean development BCE over 21 epochs for both methods and seeds. Each observation uses the same 224 development A recordings. Markers show the selected epoch for each model.

The minimum clean A BCE checkpoint is chosen, and tier C evaluates its complete development grid. The risk guards are compared as a relative comparison of the chosen checkpoints. No other epoch is chosen and no change is made to the fixed decision rule.

Figure 1 shows the clean validation curves. While all runs were completed through epoch 21, the chosen checkpoints range between epoch 13 and epoch 19. The curves represent the same validation loss on the same group of individuals. The training losses are different depending on the view policy.

4.3 Treatment, numerical contract and reproducibility

Both of the treatment arms employ the same binary labels and task of classification. An altered point of view retains its source identifier and labeling. It thus ensures equal number of steps performed by the optimizer and visits made to the source data. The clean, gain, noise and codec views are all using the same preprocessing policy. The weights of each view are individually learned in their respective arms. The estimation captures the cumulative effect of the augmentation treatment and post-registration selection, thus includes variations of the epoch selection.

The frozen normalization policy prevents recomputing the statistics from a potentially varying mixture of training images. It prevents one source of potential between-arm variability yet limits the adaptation to a new input distribution. Both the warming up and partial fine-tuning stages are conducted on both arms with the same schedule and update accounting. The optimizer update unit is the committed update. The checkpoints are storing the numerical state and source cursor for resuming an execution without altering the registration order.

During inference the checkpoint chosen according to the selection policy is retrieved from the artifact seal and receives all of the registered views according to the same P0 feature contract. P0 consists of the power floor processing, reference peaks per segment, dynamic range clipping, interpolation and channel copying. Though the latter procedure eliminates the sensitivity to uniform gains in the linear range, near floor values, padding, clipping, and interpolation do make a difference. Consistent preprocessing helps with interpretability and leaves room for exploring

the effect in other pipelines.

Every model-source-condition combination yields the score and decision for the analysis. To compute the paired differences the source and condition identifiers have to match, an absence or change of one of those rows is thus invalidating the comparison. This accounting also distinguishes the final validation stage of the selected checkpoint from the optimization monitoring process. The epoch described in the exported model may differ from the validation reported by the live monitor.

4.4 Why selection is part of the intervention

In this method, a shared policy can be enforced, yet it allows the selected epoch to be contingent upon the training policy itself. At the time of the comparison, a new optimization problem can be solved as well. The difference is that for the comparison performed there, the epoch chosen by Clean Loss Selection for each arm will remain unchanged, and the stress scores won't be taken into account. Revealing the selected epochs helps to make the distinction clear and not confuse any of them with final weights.

The source sequence is also identical to eliminate an effect of different scheduling of source arms on their observed differences. View selections by channel training include some stochasticity in contrast to the identity selection of the ERM arm. That is what differs them from each other. These two views are applied in an ordered stress chain of views, although composed views do not appear in training-view bank. That is why the improvement in terms of composition measures transfer of views within a given cohort and is conditional upon it.

5 Experimental Setup

5.1 Cohorts and conditions

For the resource-bounded medium cohort, there are 640 ASVspoof 2019 LA training utterances, evenly divided between the REAL and FAKE categories and with attack IDs A01-A06. Development data set A contains 112 utterances each of the REAL and FAKE categories, using seven speakers. Development data set B includes 144 REAL recordings, using nine speakers. The registered audit will use disjoint training and development speaker IDs. The same development data sets are used by the three independently initialized blocks, hence the blocks will measure sensitivity to training data and not a fresh test set.

Seven scored views for each model include a clean, two gain controlled, Noise5, Opus64, and the two reversed chains. The stress endpoint is made of Noise5 and chains. In chains, there is a constant-absolute 10 dB noise configuration and 24 kb/s Opus encoder in each order. Noise5 and Opus64 are distinct conditions. The corresponding standalone Noise10 and Opus24 were done on seeds 9107 and 9207. Seed 9307 will give only development order scores. Table 3 gives an overview of fixed design.

The medium sample was selected because it would allow equal CPU runs for both REAL and FAKE classes. It includes 320 REAL and 320 FAKE samples. The full training partition contains 2,580 REAL and 22,800 FAKE recordings, so a comparison with that partition would change both training size and class mixture.

Table 3: Registered prospective development experiment.

Factor	Specification
Training	640 sources, 320 per class
Development	A 224 mixed-class, B 144 REAL
Blocks	Matched development seeds 9107, 9207 and 9307
Architecture	MobileNetV3Small with LSTM128
Budget	21 epochs, 840 updates per arm
Selection	Minimum A clean BCE
Operating point	FAKE when score is at least 0.5
Outputs	2,576 selected-model scores per arm

5.2 Paired uncertainty and efficiency

For each seed, differences are channel minus ERM. Percentile intervals use 10,000 paired speaker-cluster bootstrap draws with seed 91109. A and B are sampled separately. Within a sampled speaker, all sources and their views retain their paired model scores. This preserves source and speaker dependence and avoids treating transformed views as independent trials. Mean, sample standard deviation and range across the three blocks are descriptive. They do not estimate a population confidence interval over random initializations.

All the eight fits were performed on a Windows CPU platform. The optimizer time and validation time are drawn from the run log and may include iterations after interruptions. The registered evaluation method maintains the 0.5 cut-off point without the calibration step.

5.3 Nested training-size comparison

This larger group of training has the original 640 samples and another 640 samples of the training set. This dataset has 640 REAL and 640 FAKE samples for 20 speakers and covers A01-A06. The attack counts are 106, 108, 108, 108, 106 and 104. The audit evaluated audio hash uniqueness and disjoint IDs, speaker IDs and audio hashes compared to the development, calibration (reserved), and locked out subsets. Lineage of utterances is still not complete. The two bigger limbs share the same fresh initialization that their smaller 640 peers have at seed 9207. All four models employ the same 368 development utterances, seven conditions, clean BCE checkpoint choice and threshold 0.5.

For endpoint M, the size difference will be the difference between the two size augmentation differences:

$$\begin{aligned} \Delta_s &= M_{\mathrm{Channel},s} - M_{\mathrm{ERM},s}, \\ \Delta_{\mathrm{size}} &= \Delta_{1280} - \Delta_{640}. \end{aligned} \tag{7}$$

Higher positive differences mean higher ROC-AUC ranking and higher error rate for false alarm and miss. Both cohort sizes have 21 epochs, meaning that cohort size doubling also results in doubling the budget of optimization steps from 840 to 1,680. The estimand is sensitivity under this fixed-epoch policy. We conducted 10,000 pairs of bootstrap draws for speakers per each development stratification, using the same weight values for all four models fitted and all

perspectives from the samples of speakers. Pointwise intervals capture the evaluation cohort variation given the fitted models. Training-seed and training-cohort variability are outside these intervals.

5.4 Design controls and support

Development A and B serve different purposes. A provides both classes needed for ROC-AUC, EER and fixed threshold risks. B introduces an additional stratum of REAL-only speakers that helps to measure false alarm sensitivity. Merging B into A would alter the inherent weighting of speakers as well as the set of REAL-FAKE pairs used. The two strata are kept separate. Class counts in the mixed-class stratum are provided by 112 unique recordings per class. The three stress views add up to 336 observations per class but still rely on those same recordings.

For each seed block, the two fitted models get the identical set of development sources and transformed waveforms. Between seeds, different initializations produce different fitted functions, while the evaluation cohort is identical. Thereby, we have two types of pairing. Source pairing determines the input to a comparison of the methods, while seed blocking is responsible for different initializations and source ordering in fitting. All three seed blocks use the same speaker population. All attacks A01-A06 occur in training. Hence, the attack rows represent variation in attack IDs across stress.

Order experiment places the same noise realization either before or after the codec with the same nominal clean-referenced signal-to-noise ratio. However, the achieved characteristics after compression might differ. Assuming the equality of achieved SNR or bandwidth would eliminate some part of the intervention. The controls allow comparing those particular pipeline combinations and not the simulations of packet loss, device reverberation, proprietary enhancement or communications stack altogether.

5.5 Interpretation of the resampling intervals

All observations from the sampled speakers will remain clustered [26]. Within each replicate, the speaker weights will be the same for the ERM and the channel within a stratum. If a speaker is sampled multiple times, it will contribute the repeated cluster weights to the calculation of the contrast, which then repeats its calculations based on those weights and reports the percentile limits among the valid draws. Thus, repeated views of the same source and the predictions in the paired arms remain linked through their sampling process. For rank statistics, the constraint on class support will be applied through the resampling procedure. The record of the number of valid draws allows the interval to take into account the draws that are undefined.

The intervals obtained based on seven or nine speakers only provide some empirical support to the percentile interval approach. The large effect from one speaker may alter the confidence interval and the selection of the attacks represented within the resample. The intervals are conditioned on the fitted models and, hence, do not reflect any change of the training seed. Descriptive summary is used to describe the variability between different seeds.

5.6 Evidence boundaries

The prospective comparison encompasses the registered score orientation, 0.5 operational rule, budget, clean-loss criterion for selecting check points, and the stress condition. The pair of records determines the unit in which the methods are compared. The additional Noise10 and Opus24 runs for 9107 were introduced to serve as diagnostics on the same fitted models and development sets. This helps in understanding the deviation from the additive output assumption without any independent new-cohort replication.

5.7 Locked Held-Out Evaluation

The main holdout set contains 1,088 audios from 67 speakers from the official ASVspoof 2019 LA countermeasure evaluation partition [9]. Using a deterministic hash-based ranking and a speaker round-robin with seed 91109 we have 256 REAL and 64 FAKE recordings for each attack A07 to A19. The audit was reconciled with the archive and did not find any matches between the selected subset and training, development, and calibration manifests for shared source ID, speaker ID and audio SHA-256 digest. Completeness of the generator and utterance lineage verification remains to be done.

The lock includes the four development checkpoints from 9107 and 9207 and four conditions per source: clean, Noise5, Noise10 then Opus24, and Opus24 then Noise10. All 17,408 predictions were made. Seeds 9307 and both fits with 1,280 recordings provide only the development evidence. The sigmoid score is oriented towards FAKE and the score larger than 0.5 goes to FAKE. Macro stressed ROC-AUC takes an average of the three stress-condition AUCs. The clean and stressed error rates have class-conditional denominators and the stressed denominator uses three correlated views per source. The held-out contrast holds within-seed source pairing and applies the same speaker-bootstrap to the held-out speakers. Development A and B refer to the early development strata only.

Locked sources were matched against their respective identity locks. All per-source score shards preserve full model and condition sets and have been validated prior to analysis. Registered metric values and associated uncertainties have been recomputed from committed predictions. The selected sources, models, transformations, and threshold are locked before scoring. The attack-balanced partition has distinct support from development partitions A and B; thus, differences in effect magnitude across partitions are only descriptive.

The ordered views employ source decoding to 22.05 kHz and truncation at 3 s followed by the standard peak limit of 0.25. A source-ID-keyed random selection provides a single fixed-absolute Noise10 vector normalized against clean reference power. Both chains use this vector. The codec chain uses FFmpeg with libopus at nominal rate of 24 kb/s, PCM16 input, aligned timestamps and decoding to 22.05 kHz. There is no post-channel normalization of the waveform before P0. The use of the common noise vector does not mean identical post-codec signal-to-noise ratios. Development replay verifies the composed waveforms and score anchors. Held-out evidence uses locked implementation and scores.

Table 4: Development results for the three 640-recording seed pairs. The selected update identifies the clean-BCE checkpoint. Stressed counts use repeated views.

Seed and arm	Update	Clean AUC	Stress AUC	Stress FA	Stress miss	B clean FA
9107 ERM	560	0.9872	0.6715	54/336	222/336	19/144
9107 CHANNEL	760	0.9814	0.8026	45/336	155/336	31/144
9207 ERM	520	0.9851	0.5946	167/336	143/336	26/144
9207 CHANNEL	720	0.9809	0.7839	70/336	134/336	20/144
9307 ERM	360	0.9822	0.5498	44/336	266/336	21/144
9307 CHANNEL	680	0.9856	0.7934	61/336	119/336	30/144

Table 5: Channel minus ERM. Brackets give within-seed 95 percent speaker-cluster intervals. Mean and sample SD describe the three fitted contrasts. Error differences are percentage points.

Endpoint	9107	9207	9307	Mean and SD
A macro stress AUC	+0.1312 [+0.0207, +0.2387]	+0.1893 [+0.0932, +0.2684]	+0.2436 [+0.1884, +0.3071]	+0.1880 ± 0.0562
A clean REAL FA	+0.00 [-2.68, +2.68]	+0.00 [-2.68, +2.68]	+1.79 [-3.57, +9.82]	+0.60 ± 1.03
A clean FAKE miss	+2.68 [+0.00, +6.25]	+0.00 [-3.57, +4.46]	-1.79 [-4.46, +1.79]	+0.30 ± 2.25
A stress REAL FA	-2.68 [-8.63, +4.76]	-28.87 [-37.20, -19.05]	+5.06 [-1.79, +10.42]	-8.83 ± 17.78
A stress FAKE miss	-19.94 [-25.31, -14.58]	-2.68 [-10.42, +3.57]	-43.75 [-47.92, -39.58]	-22.12 ± 20.62
B clean REAL FA	+8.33 [+1.39, +17.36]	-4.17 [-6.94, -0.69]	+6.25 [-1.39, +15.28]	+3.47 ± 6.70
A05 stress FAKE miss	+3.70 [-3.70, +12.70]	+16.67 [+3.70, +29.82]	-11.11 [-16.67, -5.26]	+3.09 ± 13.90
A06 stress FAKE miss	+7.41 [-6.67, +22.92]	+27.78 [+11.76, +43.75]	-31.48 [-43.86, -17.54]	+1.23 ± 30.11

6 Results

6.1 Overall Robustness with Channel Augmentation

Table 4 summarizes the six 640-record development curves. Macro stress ROC-AUC increases by 0.1312, 0.1893 and 0.2436 for 9107, 9207 and 9307, respectively. The mean and sample SD across seeds are 0.6053 ± 0.0616 for ERM, 0.7933 ± 0.0094 for channel, and +0.1880 ± 0.0562 for their paired difference. The clean AUC difference is negative in the first two seeds and positive in 9307. Figure 2 links the two arms within each seed.

Clean A REAL false alarms tie in the first two pairs and increase from 4 to 6 of 112 in 9307. Clean A FAKE misses increase from 6 to 9 in 9107, tie at 9 in 9207, and decrease from 10 to 8 in 9307. Clean B REAL false alarms increase in 9107 and 9307 and decrease in 9207. Table 5 separates within-seed speaker intervals from descriptive variation across fitted pairs.

All three 640-recording development A macro stress AUC intervals exclude zero. In 9307 the paired interval is [0.1884, 0.3071]. The 9207 stress miss interval includes zero. These intervals quantify speaker resampling conditional on each fitted pair and shared development protocol.

In the nested seed-9207 comparison, the channel-minus-ERM macro stress AUC gain decreases from +0.1893 at 640 recordings to +0.1081 at 1,280 (Table 7). Equation (7) gives a difference of −0.0812, with a pointwise 95% paired speaker interval of [−0.1421, −0.0148]. Both fitted pairs gain in ranking. The gain is smaller under the larger fixed-epoch protocol, which also uses twice

Table 6: Development A by condition. Each cell gives ROC-AUC / REAL false alarms / FAKE misses. Each error count has 112 sources of its class.

Condition	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Clean	0.9872 / 3 / 6	0.9814 / 3 / 9	0.9851 / 4 / 9	0.9809 / 4 / 9
Half gain	0.9812 / 0 / 15	0.9818 / 0 / 14	0.9758 / 0 / 16	0.9820 / 0 / 15
Quarter gain	0.9237 / 7 / 20	0.9576 / 0 / 17	0.9278 / 9 / 20	0.9590 / 0 / 19
Noise5	0.6567 / 1 / 99	0.7793 / 8 / 64	0.5449 / 22 / 77	0.7579 / 13 / 56
Opus64	0.9884 / 6 / 4	0.9819 / 4 / 6	0.9861 / 9 / 5	0.9839 / 5 / 8
Noise to Opus	0.6763 / 48 / 38	0.8108 / 28 / 37	0.6408 / 104 / 13	0.7981 / 44 / 26
Opus to Noise	0.6814 / 5 / 85	0.8178 / 9 / 54	0.5981 / 41 / 53	0.7958 / 13 / 52

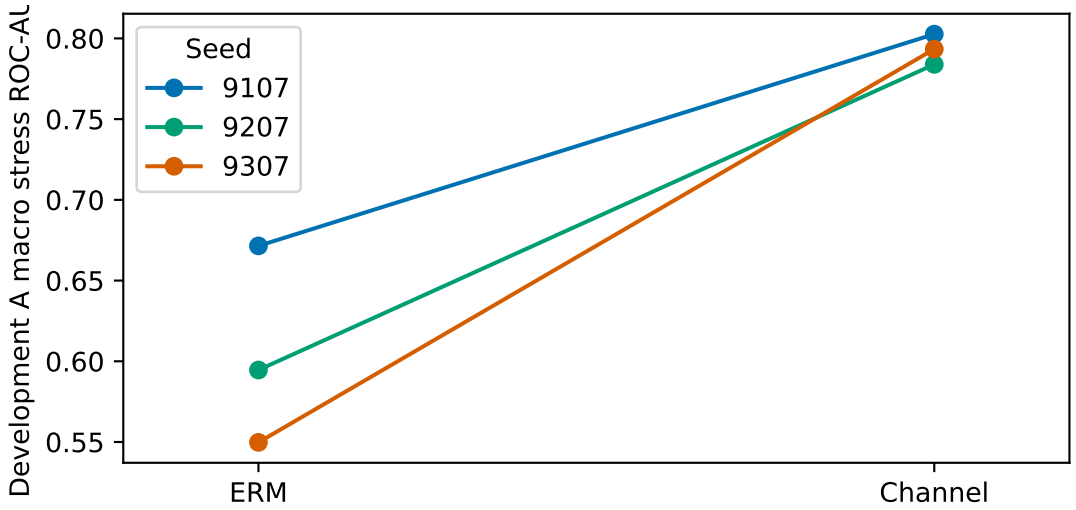


Figure 2: Development macro stress ROC-AUC for the three initialization blocks. Lines connect the two arms within a seed.

as many optimizer updates.

The locked evaluation completed 17,408 predictions for the original four selected models. Table 8 reports aggregate results and Table 9 separates the conditions. Macro stressed ROC-AUC increased in both seed blocks, from 0.8308 to 0.8875 in 9107 and from 0.7498 to 0.8948 in 9207. The corresponding paired speaker-cluster intervals for channel minus ERM were [0.0296, 0.0842] and [0.1158, 0.1752]. These positive differences were smaller than the development A contrasts. Figure 4 compares the predefined endpoints across partitions. Figure 5 shows the score-level ranking under each held-out condition.

Stressed FAKE misses fell from 1,370 to 587 in 9107 and from 815 to 453 in 9207, each out of 2,496 views from 832 FAKE sources. The rate differences were −31.37 points [−33.41, −29.36] and −14.50 points [−17.38, −11.67]. Stressed REAL false alarms increased by one in 9107, from 80/768 to 81/768 (+0.13 points), and fell from 322/768 to 136/768 in 9207. The 768 views in each comparison came from 256 REAL sources. The paired intervals were [−3.23, +3.76] and [−29.07, −19.03] percentage points. All endpoint and attack intervals used 10,000 valid draws.

Table 7: Fixed-epoch training-size sensitivity for seed 9207 on the same development sources. Column headings give the training size. A has 112 sources per class and B has 144 REAL sources. Pooled stress counts use three views per source at threshold 0.5. Each attack row uses 18 sources.

Development endpoint	ERM 640	Channel 640	ERM 1,280	Channel 1,280
A clean ROC-AUC	0.9851	0.9809	0.9900	0.9860
A macro stress ROC-AUC	0.5946	0.7839	0.7062	0.8143
A clean REAL false alarms	4/112	4/112	1/112	3/112
A clean FAKE misses	9/112	9/112	3/112	8/112
A stress REAL false alarms	167/336	70/336	5/336	161/336
A stress FAKE misses	143/336	134/336	248/336	64/336
B clean REAL false alarms	26/144	20/144	18/144	22/144
A05 stressed FAKE misses	38/54	47/54	50/54	36/54
A06 stressed FAKE misses	16/54	31/54	50/54	11/54

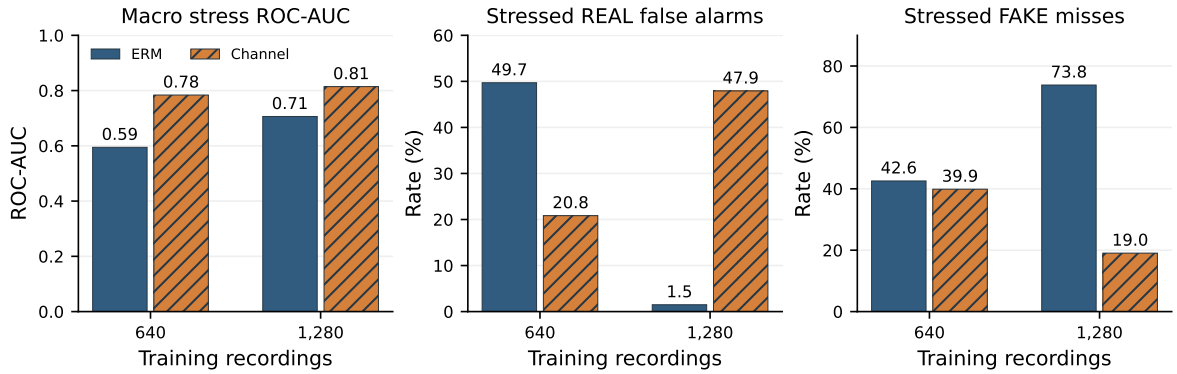


Figure 3: Seed-9207 development A results on the same recordings at both training sizes. Macro stress ROC-AUC averages three condition-level AUCs. Each pooled error rate uses 336 views from 112 sources of its class at threshold 0.5. Bars show observed values. Table 7 gives counts and the text reports the paired intervals.

6.2 Class-Asymmetric Errors and Operating-Point Trade-offs

With 1,280 samples, Channel improves stressed FAKE misses from 248 to 64 out of 336 (−54.76 points) and increases stressed REAL false alarms from 5 to 161 out of 336 (+46.43 points). With 640 samples, the respective contrast for REAL false alarms is −28.87 points, leading to a size contrast of +75.30 points (pointwise 95% speaker interval [62.50, 87.80]). Figure 3 illustrates that better ranking is accompanied by switching the class of errors at the threshold 0.5.

The selected 1,280 Channel checkpoint violates both registered +2-point clean-risk guards. Development A clean FAKE misses rose from 3/112 to 8/112 (+4.46 points) and B clean REAL false alarms rose from 18/144 to 22/144 (+2.78 points). Pointwise 95% speaker intervals were [1.79, 6.25] and [−4.86, 11.81] percentage points. The B interval includes zero. The guards use the observed differences for the pair selected by clean BCE, so both exceed the tolerance.

Table 6 breaks down the parts of the macro stress outcome. The table and ROC figure detail the first two seeds. The channel arm also increases AUC under all registered stress conditions in 9307. Figure 7 depicts the complete ROC curves associated with these AUCs. Fixed-threshold changes remain heterogeneous. Under Noise5, FAKE misses fall from 99 to 64 in 9107 and from 77 to 56 in 9207. The corresponding REAL false-alarm counts rise from 1 to 8 in 9107 and fall from 22 to 13 in 9207.

Table 8: Selected-model results on the locked held-out subset. Stress error counts pool three views per source. Macro stress ROC-AUC averages the three condition-specific AUCs.

Endpoint	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Macro stress ROC-AUC	0.8308	0.8875	0.7498	0.8948
Stress FAKE miss	1370/2496	587/2496	815/2496	453/2496
Stress REAL false alarm	80/768	81/768	322/768	136/768
Clean REAL false alarm	25/256	19/256	22/256	16/256
Clean FAKE miss	125/832	129/832	127/832	147/832

Table 9: Condition-specific held-out performance. Each cell gives ROC-AUC / REAL false alarms / FAKE misses. There are 256 REAL and 832 FAKE recordings per condition.

Condition	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Clean	0.9534 / 25 / 125	0.9436 / 19 / 129	0.9507 / 22 / 127	0.9410 / 16 / 147
Noise5	0.8053 / 0 / 701	0.8655 / 12 / 259	0.6815 / 28 / 492	0.8765 / 25 / 193
Noise to Opus	0.8447 / 76 / 143	0.8991 / 53 / 143	0.8196 / 226 / 38	0.9044 / 89 / 93
Opus to noise	0.8424 / 4 / 526	0.8979 / 16 / 185	0.7483 / 68 / 285	0.9034 / 22 / 167

The larger 9207 REAL benefit under the two compositions is visible in the per-condition counts. Noise-to-Opus false alarms fall from 104 to 44, while its FAKE misses increase from 13 to 26. By way of contrast, REAL false alarms go from 41 down to 13, while FAKE misses go from 53 down to 52. Both counts have denominator 112. The starting point with the high number of REAL false alarms for the condition of Noise-to-Opus allows a sizable decrease in this operating point. It would not be accurate to refer to the decrease itself as an advantage across classes.

For the stress results, gain controls and Opus64 for the two development seeds give some perspective. For quarter gain, the channel arm has fewer false alarms and higher AUC in both blocks. For standalone Opus64, the channel arm has fewer false alarms but higher FAKE misses in both blocks. The control of 64 kb/s uses a different rate from the compositions of 24 kb/s. It is listed as a separate condition, and any comparison against the chains would alter the codec and noise conditions as well. The 24 kb/s control is the correct one for standalone comparison against 9107 and 9207.

In 9307, stress FAKE misses fall from 266 to 119 of 336 while REAL false alarms rise from 44 to 61 of 336. This contrasts with fewer stressed REAL false alarms in 9107 and 9207. The 9307 stress miss contrast is -43.75 percentage points (95% speaker interval $[-47.92, -39.58]$). The contrast for its stress REAL false alarm is $+5.06$ points (interval $[-1.79, +10.42]$). B clean false alarms increase from 21 to 30 out of 144, which is a change of $+6.25$ points (95% speaker interval $[-1.39, +15.28]$). The difference is greater than the registered $+2$-point guard for the selected CLEAN_BCE pair. The interval represents speaker resampling in the pair and does not affect the observed guard choice.

In 9207, decreases in other attack instances balance 24 extra A05 and A06 missed instances under stress conditions. The A05 and A06 attacks add six extra missed instances in 9107. The differences include transformed views. Table 11 gives the number of A05 and A06 attacks that get an additional correct or missed decision in at least one stress condition. Figure 8 provides the paired analysis for both classes under A.

Clean REAL false alarms fell from 25 to 19 in 9107 and from 22 to 16 in 9207, out of 256

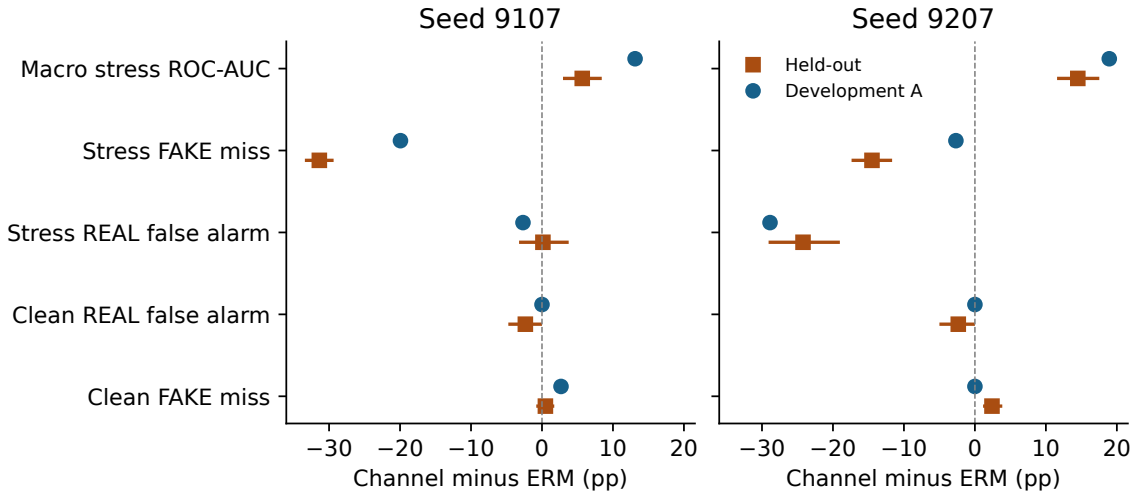


Figure 4: Channel-minus-ERM contrasts on development A and the locked held-out subset, separately by seed. Held-out segments show 95% paired speaker-cluster percentile intervals. AUC differences are multiplied by 100. Positive differences indicate better AUC or higher error rates. Cohorts and attack populations differ across partitions.

sources per arm. Clean FAKE misses increased from 125 to 129 and from 127 to 147 out of 832, while clean ROC-AUC declined from 0.9534 to 0.9436 and from 0.9507 to 0.9410. The 9207 clean FAKE miss increase was 2.40 percentage points [1.15, 3.86]. Development B showed opposite false-alarm directions between the two seeds. The cohorts differ in speaker and source support.

The sums for stress reflect the existence of the operating point shift. For Noise5, the misses of FAKE dropped from 701 to 259 and from 492 to 193 out of 832. With respect to Opus and then Noise10, the misses of FAKE dropped from 526 to 185 and from 285 to 167. In case of the reverse sequence, Noise10 followed by Opus, the misses of FAKE remained constant at 143 in 9107 and increased from 38 to 93 in 9207, while the number of REAL false alarms decreased from 226 to 89 out of 256 and AUC increased from 0.8196 to 0.9044.

Post hoc analysis shows that at the threshold of 0.5 the operating point has shifted. When Noise10 and then Opus were applied in 9207, the two algorithms agreed on 85 real false alarms, channel corrected 141 and made four. In case of FAKE, they agreed on 20 misses, channel corrected 18 and introduced 73. In 9107, when the two algorithms were subjected to Noise10 and then Opus, the sum of 143 FAKE misses included 93 shared misses, 50 corrected and 50 introduced. The numbers reflect decision-making on the same sources.

In Fig. 9, the retro-grid ensures that the registered threshold remains at 0.5. On held-out 9207 under Noise10 followed by Opus24, channel augmentation gives fewer false alarms and more misses at every sampled threshold from 0.20 through 0.80. The miss direction reverses at 0.10 and 0.90. For Noise5 and the opposite order, FAKE misses decrease at all 11 sampled thresholds in both seeds, with condition- and seed-dependent REAL errors. FAKE misses cleaned in 9207 rise for all thresholds sampled. These comparisons describe the measured score grid and provide no threshold-selection rule.

Score distributions also change locally around the registered boundary. Under Noise10 followed by Opus24 in 9207, 7 versus 30 of 256 REAL scores and 5 versus 18 of 832 FAKE scores lie within 0.025 of 0.5 for ERM and channel, respectively. The supplement gives the

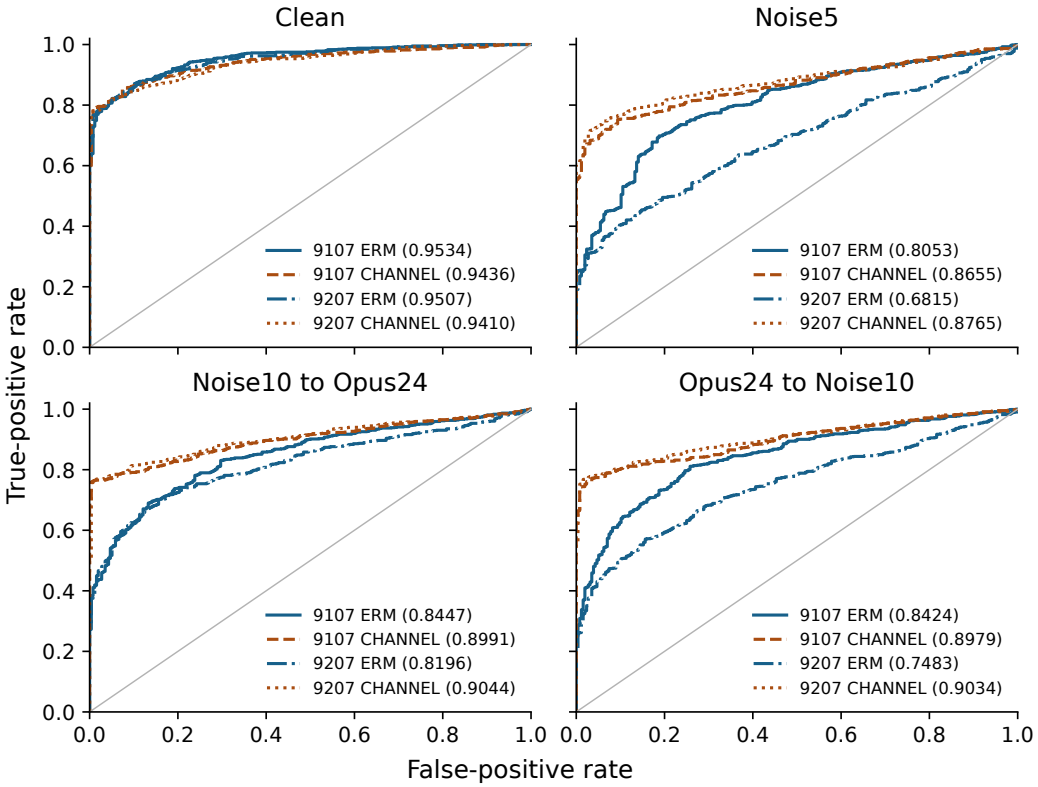


Figure 5: Empirical held-out ROC curves for four selected models. Each panel contains 256 REAL and 832 FAKE recordings, evaluated under the named condition. Legends give seed, method, and AUC. The same sources recur across conditions.

confusion matrices for every arm and condition, empirical curve coordinates and the 0.025 and 0.05 score windows. Development B supplies only REAL scores. Its FNR, ROC-AUC and EER are undefined.

6.3 Attack-Specific Robustness and Failure Modes

For seed 9207, the A05 contrast changes from 38/54 versus 47/54 at 640 to 50/54 versus 36/54 at 1,280 (ERM versus Channel). A06 changes from 16/54 versus 31/54 to 50/54 versus 11/54. Each denominator represents three correlated stress views from 18 sources. Both training size and update budget changed. Seed 9307 compares initializations at 640. Table 10 and Fig. 10 show both sign reversals. Channel increases A05 and A06 stress misses in the first two pairs and decreases them in 9307. The latter changes are $-6/54$ and $-17/54$, respectively. The pointwise 9307 speaker intervals are $[-16.67, -5.26]$ and $[-43.86, -17.54]$ percentage points, respectively. The 9107 intervals include zero. These within-seed intervals do not establish a stable attack effect across initializations.

Table 11 details the first two seeds. In 9307, all six A05 recoveries occur in the noise-before-Opus condition. For A06, 18 recoveries and one new miss occur across the three stress views. Net recoveries span five A05 speakers and six A06 speakers. The A06 ERM burden is 16 of 54 in 9207 and 45 of 54 in 9307. Differences in baseline burden and FAKE score crossings describe where the reversal occurs. They leave the underlying model or generator mechanism unresolved.

Figure 11 shows that two B speakers contribute 10 of the 12 net additional false alarms in

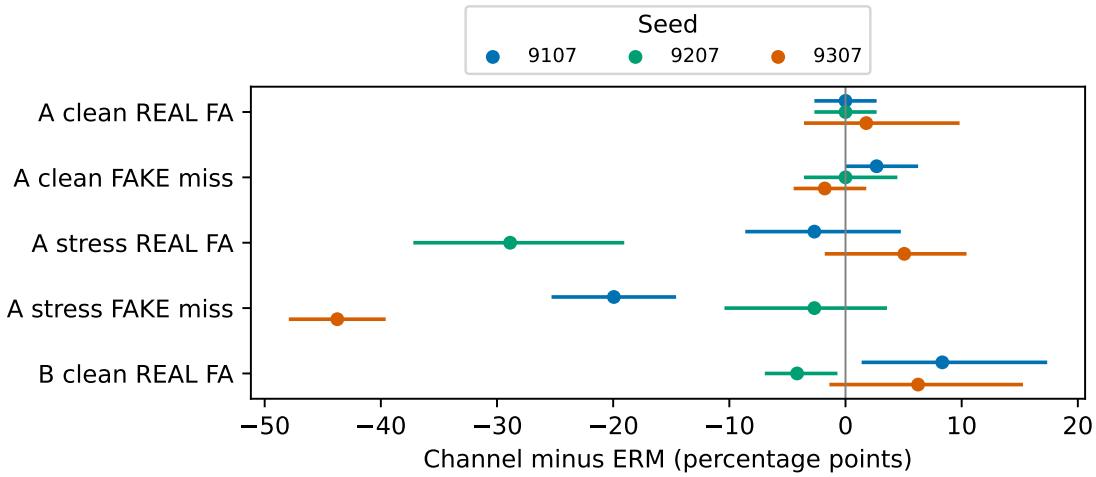


Figure 6: Channel-minus-ERM fixed-threshold error rates by development stratum, class, and seed. Segments show within-seed paired speaker-cluster percentile intervals. REAL errors are false alarms and FAKE errors are misses. Clean A has 112 recordings per class, stress A has 336 views from 112 recordings per class, and clean B has 144 REAL recordings.

Table 10: Development A05/A06 stress FAKE misses over the three stress views. Each row represents 18 distinct sources and 54 transformed views.

Seed and attack	ERM	Channel	Δ misses
9107 A05	48/54	50/54	+2
9107 A06	41/54	45/54	+4
9207 A05	38/54	47/54	+9
9207 A06	16/54	31/54	+15
9307 A05	49/54	43/54	-6
9307 A06	45/54	28/54	-17

9107. For 9207, reductions occur across several speakers. B contains only REAL recordings, and the counts are interpreted as false alarms with no B ROC-AUC or EER.

Figure 12 shows the condition-specific FAKE miss contrasts. Pooled stressed misses fell on A07–A16 and rose on A17–A19 in both blocks. A17, A18 and A19 added 10, 11 and 4 misses in 9107 and 43, 43 and 56 in 9207, each out of 192 views from 64 sources per attack. The pointwise 9107 intervals include zero. The 9207 intervals exclude zero: [15.90, 29.03], [14.12, 30.81] and [20.63, 37.70] percentage points. The development A05/A06 results concern a separate attack set.

In the post-hoc grouping, A07–A16 account for 808 and 504 fewer stressed misses in 9107 and 9207, partly offset by 25 and 142 additional misses on A17–A19. For A17–A19, Noise-to-Opus added 39 misses in 9107. All three stress conditions added misses in 9207. On A19, 43/64 sources acquired a miss in at least one stress condition and 18/64 in multiple conditions. These exploratory groupings were identified after reviewing held-out results. The pointwise intervals have no multiplicity adjustment.

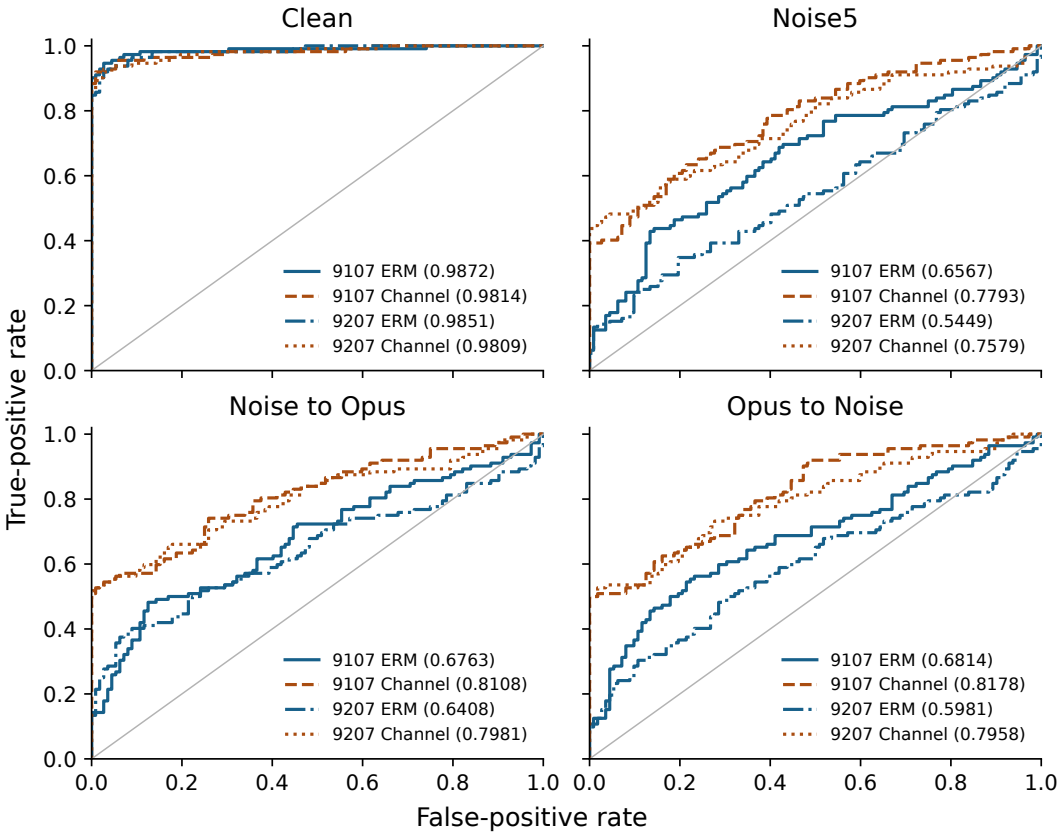


Figure 7: Empirical ROC curves from the four selected models in the first two seeds on development A. Each panel ranks the same 112 REAL and 112 FAKE recordings under one condition. Legends give the seed, method, and recomputed AUC. The diagonal denotes chance ranking.

6.4 Transformation-Order Effects

For the same 112 A REAL recordings at seed 9207, the Channel-minus-ERM contrast in the signed order error gap (Noise-to-Opus minus Opus-to-Noise) changes from −28.57 points at 640 to +26.79 points at 1,280. The size contrast is +55.36 points (pointwise 95% speaker interval [32.12, 79.46]). The change in the corresponding signed FAKE-score difference is −0.2099 to +0.0764. These paired contrasts measure detector response to the order of processing. See the supplement for the counts exclusive and inclusive to each arm and class. In the first two development blocks, out of 112 A REAL sources, the Noise-to-Opus-only false-alarm counts change from 43 to 19 in 9107 and 63 to 31 in 9207. The opposite order has no exclusive REAL false alarms in A. Among 112 A FAKE sources, Opus-to-Noise alone causes 48 versus 17 misses in 9107 and 40 versus 26 in 9207. The Noise-to-Opus-only FAKE counts are 1 versus 0 and 0 versus 0. In the 9307 pair, there is also the same order asymmetry classwise. Noise-to-Opus-only REAL errors have shifted from 37 to 44, while Opus-to-Noise-only FAKE errors have shifted from 41 to 33, both in a sample of 112. The signed FAKE score gap decreases from 0.2375 to 0.0538 in the median. Figure 13 represents paired score differences before the thresholding point.

9107 retrospective diagnostics produce 1,472 predicted values for Noise10 and Opus24 separately. Matched waveforms help recover not only registered sequences but also selected anchor values that were replayed without drifting. A leave-one-speaker-out nuisance regression yields

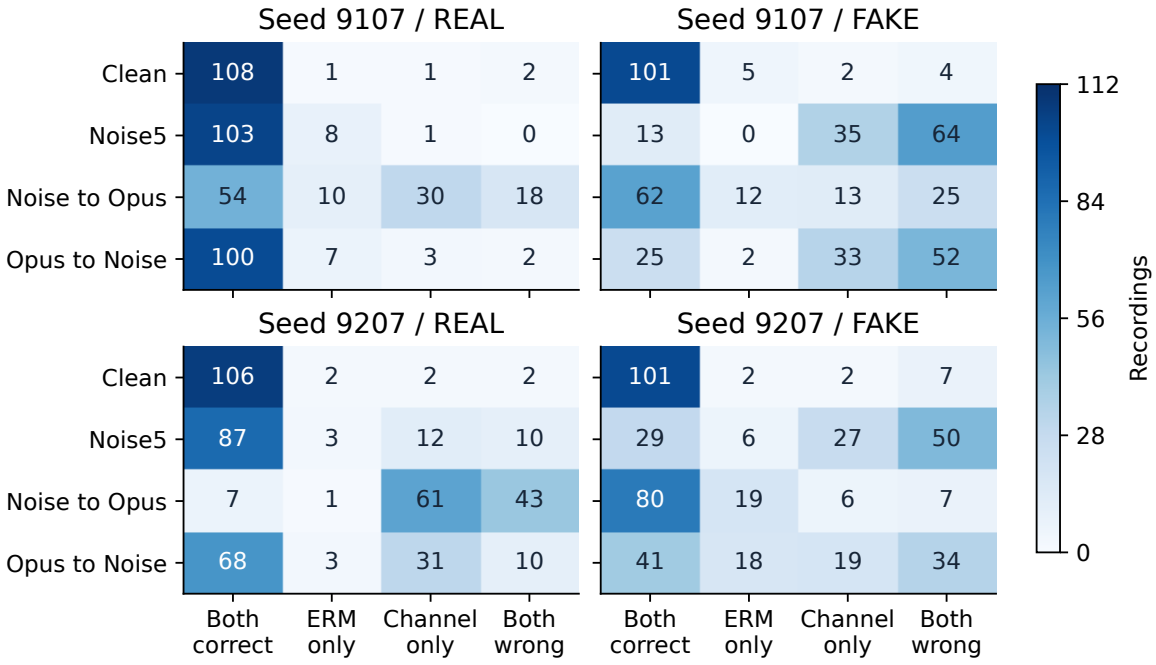


Figure 8: Within-source correctness for the first two seeds at threshold 0.5 on development A. Each row contains 112 recordings of the indicated class. The same recordings contribute to each condition row. The two middle columns identify sources correctly classified by only one method.

Table 11: A05 and A06 source-level diagnostics. Each condition has 18 sources. Error cells are ERM / channel. Harm and recovery can overlap across views.

Seed / attack	Clean	Noise5	Noise to codec	Codec to noise	All stress	Harmed / rec.
9107 / A05	0 / 0	17 / 17	14 / 16	17 / 17	14 / 16	3 / 1
9107 / A06	3 / 6	18 / 16	7 / 12	16 / 17	6 / 12	7 / 2
9207 / A05	0 / 0	17 / 17	7 / 13	14 / 17	7 / 13	8 / 0
9207 / A06	5 / 7	9 / 12	0 / 5	7 / 14	0 / 5	10 / 0

directional class-discrimination AUCs of 0.538 for ERM and 0.432 for channel on the signed order score. This measures a residual score contrast on the development cohort. The same standalone components were evaluated for seed 9207 on all 368 development sources, adding 1,472 predictions. All 18 clean and ordered score anchors matched their sealed values exactly. On development A, standalone Noise10 FAKE misses decreased from 82 to 54 in 9107 and changed from 47 to 48 in 9207, out of 112 per arm. Under standalone Opus24, 9207 REAL false alarms decreased from 19 to five and FAKE misses increased from two to ten. The 9107 nuisance regression remains a single-seed analysis. These component results constrain interpretation of the compositions and do not identify a forensic mechanism.

Source-linked order asymmetry persisted. In 9107, REAL sources wrong only under noise-to-Opus numbered 72 for ERM and 37 for channel, with zero exclusive errors in the reverse direction. In 9207 the corresponding counts were 158 and 68, while exclusive reverse-order errors numbered zero and one. Among FAKE sources, reverse-order-only misses numbered 384 and 43 in 9107 and 247 and 74 in 9207. The opposite exclusive FAKE miss counts were one and one in 9107, and zero and zero in 9207. Each model uses the same 256 REAL and 832 FAKE sources in the order comparison.

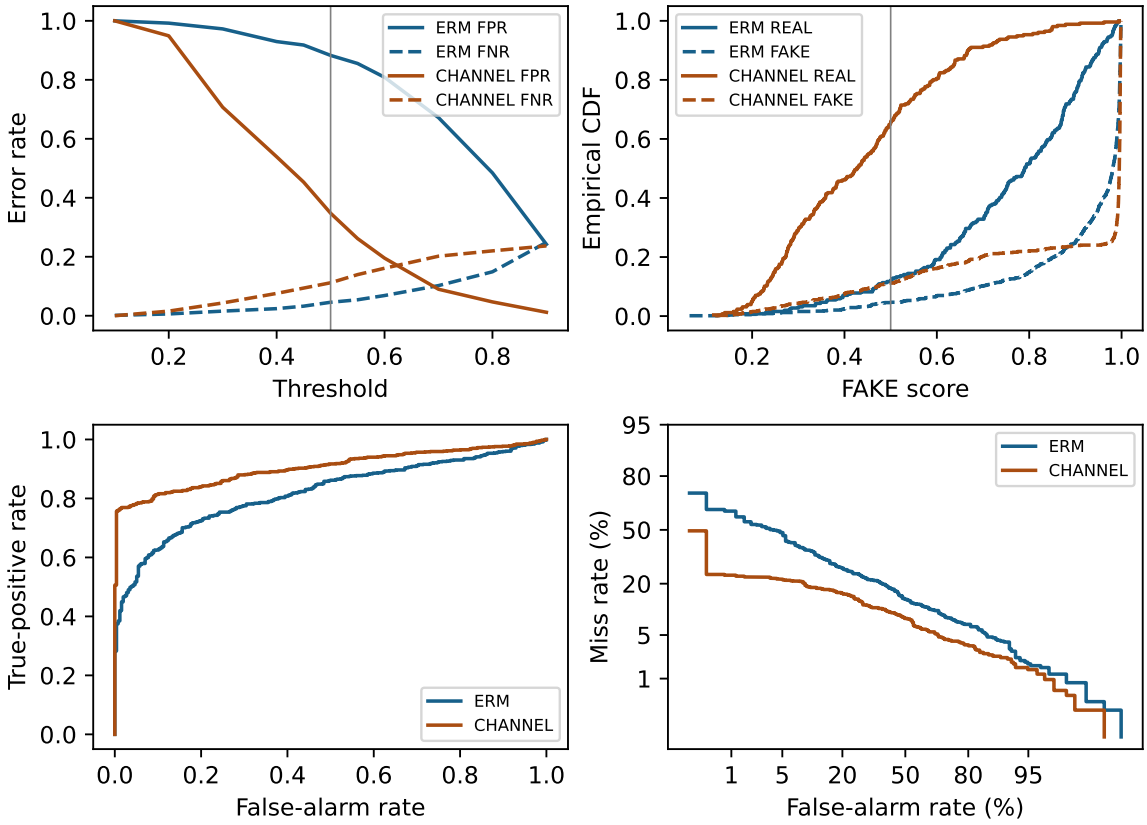


Figure 9: Held-out seed 9207 under Noise10 followed by Opus24. Top left shows exploratory FPR and FNR at 11 thresholds. Top right shows class-specific empirical CDFs. Bottom panels show empirical ROC and DET. DET axes use a normal quantile transform and clip endpoint rates at half an event per class. The vertical lines identify the registered threshold of 0.5. Each arm has 256 REAL and 832 FAKE sources.

The score contrasts shown in Fig. 14 were also directional, with reduced median FAKE scores following channel adaptation. The median score contrasts in FAKE became 0.3454 and 0.0063 in 9107, 0.2419 and 0.0076 in 9207. However, small score contrasts cannot automatically be assumed to reflect a superior decision rule, in view of the fact that attack-specific misses occurred. Speaker-related scores exhibit the same order error pattern distribution. In the case of ERM 9107, the reverse-minus-forward FAKE error contrast was 16 out of 21 FAKE recordings for speaker LA_0045 and 13 out of 21 for speaker LA_0046.

6.5 Robustness, Stability and Generalization

The direction of the development stress ranking gain and FAKE miss reduction is consistent across all three pairs. The magnitude of the gain, as well as the direction of false-alarm results, varies according to the selected pair. Common development data allow for a comparison of initializations using the protocol described above, in which case speaker uncertainty is computed in each pair separately.

The fixed epoch size experiment includes a development-only comparison for seed 9207. The three initializations correspond to 640 recordings. The held-out evaluation evaluates the original 640 recording model corresponding to seeds 9107 and 9207 using additional ASVspoof 2019 LA

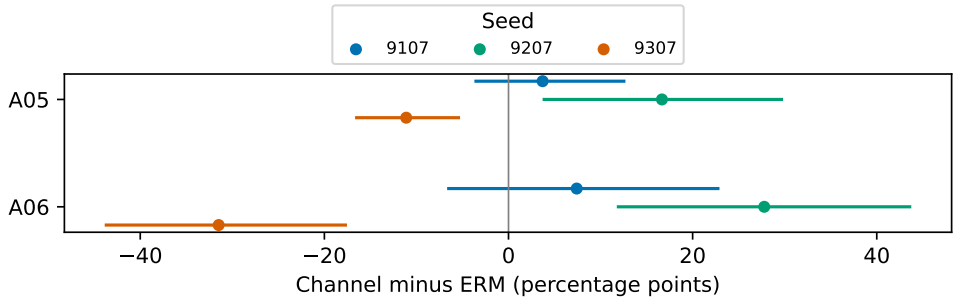


Figure 10: Development A05/A06 stressed FAKE miss changes by seed. Positive values indicate more misses. Intervals resample speakers within a seed. Each attack contributes 54 views from 18 recordings.

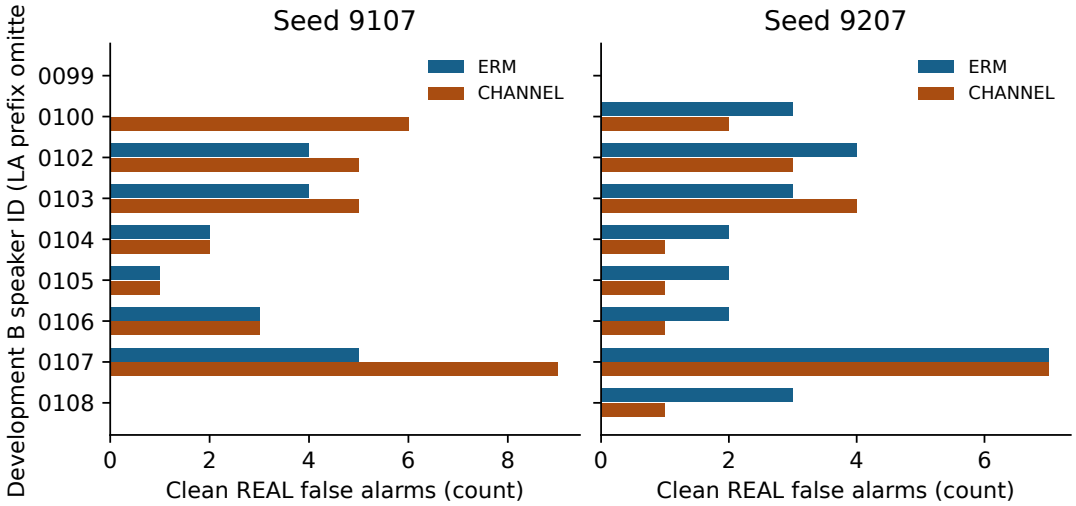


Figure 11: Clean REAL false alarms by speaker on development B for the first two seeds. Each of nine speakers contributes 16 recordings, for 144 recordings per arm and seed.

speakers. Cross-corpus robustness remains untested.

6.6 Practical Implications and Limitations

In 9307 the recorded optimizer durations were 39.9 minutes for ERM and 78.9 minutes for channel. At 1,280, the seed-9207 optimizer durations were 78.6 and 160.3 minutes, respectively, with validation taking 37.6 and 37.9 minutes. These are local CPU workload measurements. The architecture and parameter count are shared across training sizes.

A bounded inference benchmark used the clean inputs of the first 12 registered development sources, repeated three times after anchor replay, at batch size one on a single CPU thread. Input duration was capped at 3 s. Median preprocessing time was 10.9 ms (interquartile range 10.3–11.2 ms), forward time 91.3 ms (90.5-93.5 ms), and end-to-end time 102.1 ms (101.1-104.2 ms). The timed preprocessing includes decoding, resampling and the P0 input transform on clean input. The separate headroom and channel-transform stages are outside the timed path. The two 9207 exports each contain 1,308,401 parameters, of which 369,281 are updated during warm-up and 646,481 during partial fine-tuning. The exported stage has 661,920 nontrainable parameters. Each export occupies 5,916,699 bytes. Loading time in one process was 3.35 and

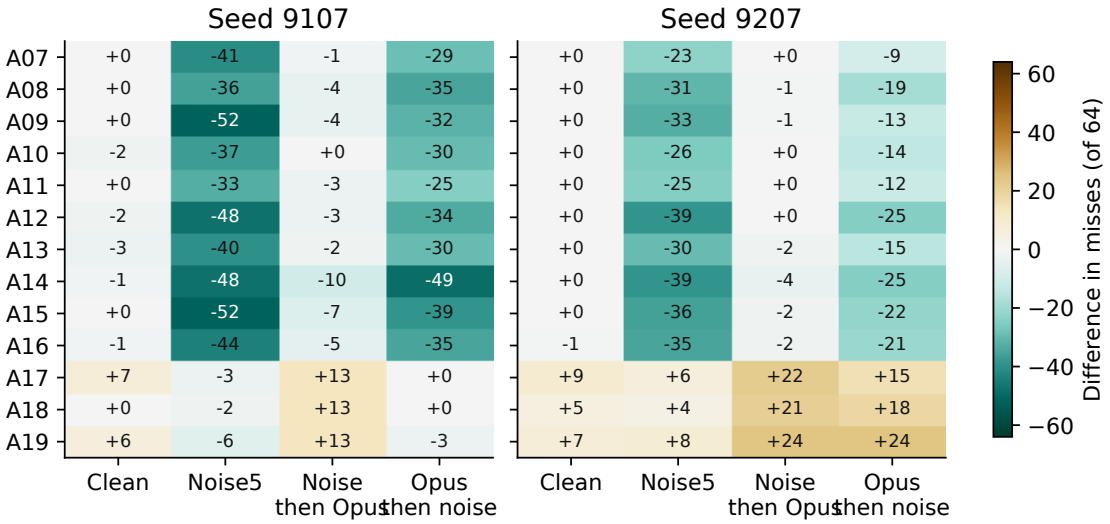


Figure 12: Held-out channel-minus-ERM FAKE miss counts by attack, condition and seed. Each cell compares the same 64 sources. Positive counts indicate deterioration and negative counts indicate fewer misses. The full numerical supplement retains both arm counts and stressed attack intervals.

2.81 s, where the second loading came after the first loading. The maximum process working set during the timing process was about 475 MiB. The development convenience sample supports an architectural and local runtime description. Mobile, streaming and concurrent performance remain unmeasured.

The error curves illustrate why a new operating point requires independent validation with costs and traffic prevalence appropriate to the deployment. The ordered chains make it more imperative to validate the position where noise is added vis-a-vis codec operations.

7 Discussion

Higher stressed ranking and fewer pooled stressed FAKE misses are observed across all three development seed blocks. A05 and A06 provide an example of how the aggregate pattern can coexist with the attack effects which are opposite for the fitted pairs. Therefore, the third block helps to support further the aggregate contrast under the registered protocol and restricts the claim about the attack-level behavior.

Clean B false alarms rise in 9107 and 9307, and fall in 9207. Stressed REAL false alarms also reverse the trend in 9307. The described effects characterize decisions under the fixed threshold. They make it necessary to report two separate risks for each class when ranking works.

The nested comparison shows why ranking and fixed-threshold errors must be read together. At 1,280, fewer stressed FAKE misses accompany more stressed REAL false alarms and two clean-risk guard failures on the clean-BCE-selected pair. Table 7 gives the counts. The smaller ranking gain accompanies the larger cohort and additional optimizer updates under the fixed-epoch protocol.

Noise-to-Opus is the sole erroneous order for many REAL sources. For FAKE sources, the inverse sequence is commonly the sole erroneous order. A rise in the FAKE score in one of the

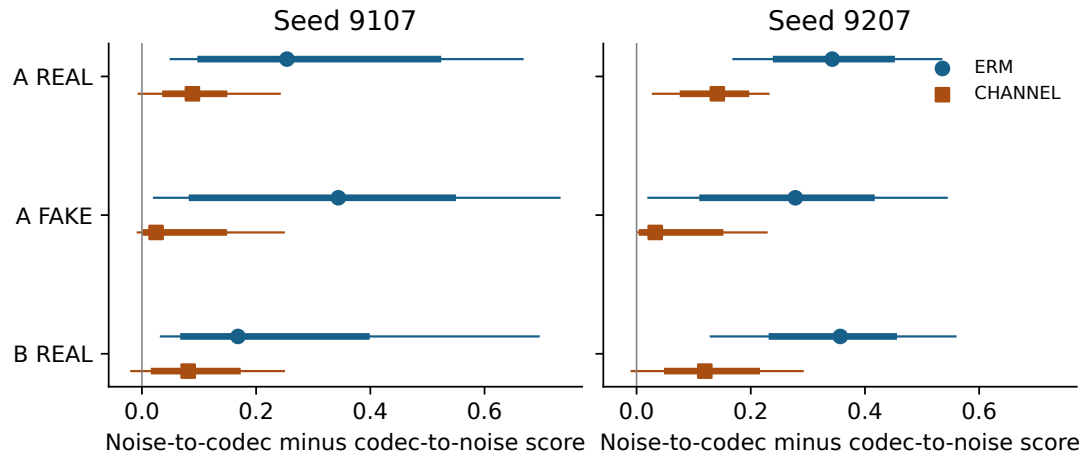


Figure 13: Directional score contrasts for the first two development seeds on the same recording in opposite processing orders. Points are medians, thick segments span quartiles, and thin segments span the 10th to 90th percentiles. Each A class has 112 sources and B has 144 REAL sources. Segments describe score distributions.

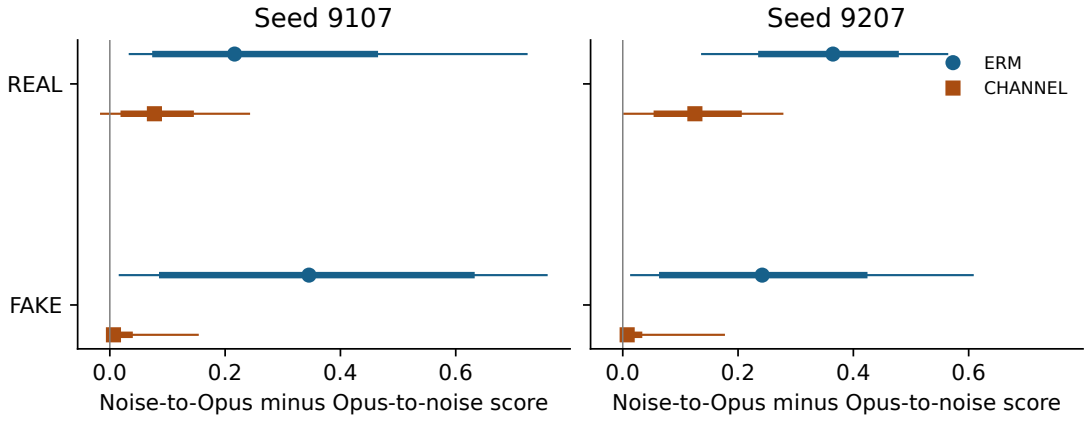


Figure 14: Held-out same-source FAKE score contrasts between noise-to-Opus and Opus-to-noise. Points are medians, thick segments span quartiles, and thin segments span the 10th to 90th percentiles. REAL and FAKE groups contain 256 and 832 sources. These are distribution summaries of paired scores.

orders may create both sequences. The retrospective matched-component diagnostic for 9107 finds that nuisance adjustment does not establish an independent forensic target. Nonlinear codec response, preprocessing, and detector sensitivity may all contribute. Reproducing the direction of decision asymmetry in 9207 and 9307 does not independently reproduce the nuisance analysis.

In practical evaluation, reporting both threshold-specific risks and ranking separates several failure modes that may have different costs. The speaker's focus is also another factor influencing a potential change in review workload, but no measurement was made for field workload or utility. Conventional Opus order reversal, class-wise counts and a matched training intervention provide a focused complement to broader robustness benchmarks and codec-adaptation methods.

7.1 Aggregate improvement and its distribution

Matched comparison is used here to make the training-view approach the independent variable, but it does not single out any individual cause in the network that was trained. Augmented channel model could learn to discriminate more pairs in terms of stress processing, shift the score distribution with respect to the static decision boundary, or even both. Clean BCE could select a different epoch for each arm because the score distributions in the network are different. In this way, the contrast observed will incorporate the effect of augmentation in combination with the identical selection method.

Both A05 and A06 have 18 sources of development, and a single source may have three error types associated with the stress. 9307 source-based changes include many speakers under many different conditions during attacks. A greater miss burden in baseline provides for more recoveries, yet the information about the scores and speaker numbers is not enough for isolating the source of seed-related flip.

FALSE positives might move around in response to a score difference which brings real cases near 0.5 across the threshold. In B there are three blocks of different directions, and some portion of 9107 increment falls on a few speakers. Bona fide cross-validation is a justification for evaluation of bona fide diversity [3]. Our speaker evaluation covers a narrower question within one static sample. A different corpus of bona fide data, different language, noise type or deployment history might alter both the baseline figures and the direction of the intervention effect.

7.2 Signal-chain and engineering implications

Published communication benchmarks like ADD-C have varying codec and packet loss characteristics [2]. This range makes them valuable as a source of coverage under stress conditions. The same-source ordered comparison presented in this study takes range out of the equation and uses strict experimental control in favor of it. It is an examination of two precisely defined chains of events under consistent settings and notes down in what order failure occurs for a certain choice to be made. Such a correlation can drive further experiments using real-life transmission with different gain levels.

In cases where review is made on a threshold basis, the deployment report should state the sequence and configuration of the processing chain, as well as error rates by class. The rank statistic will allow the assessment of the sequence of detectors, while the error transition matrix will indicate the presence of a possible transition of errors into another category. The latter information is important even when there is equality in counts. For instance, two approaches may register the same amount of valid sources, yet they require review for different records. There was no application-dependent cost measurement performed, so the cost-optimal operational criterion is another issue.

The current controlled experiment employs common BCE and one perspective per source. The scientific contribution of the experiment is the error analysis and the demonstration that the rank effect benefit is accompanied by subgroup harm. Extension of the hypothesis would entail a locked independent division, followed by a new corpus or model for measuring increased transportation. For the forensic mechanism to be confirmed, a cue definition and an independent

variable are also needed.

7.3 From a validation metric to a validation workflow

The table of conditions provides information for a situation where the engineer is weighing options regarding augmentation. Where there is an increase in the AUC value and decreases in errors of two types, then there is improvement in both levels that were considered. There is an increase in the AUC value and decrease in false alarms and increases in the number of misses. This is because the rank improvement and trade-off of the operating point occur together. A case where the number of sources is tied may still hide improvements in some sources.

Condition name labels may similarly confound the processing transformation. In naming a noise and Opus label, no information is provided about the presence of the noise during the encoding stage. This aspect is addressed with the operator labels, and the realization of the noise is defined in the process. This allows sensitivity only to the stated order without making any assumptions about either chain containing all communications routes. It further justifies the reporting of coding rates along with the order. Opus64 uses different settings and has support not included with the 24 kb/s compositions, and hence their comparison will not isolate the bitrates. The current data relates to the classifier after the stated transformation.

7.4 Evidence needed to extend the claim

Similar issues have been noted by the reviewers of the conference with respect to broader robustness and reproducibility. This work advances by isolating the training initialization step from the test-source resampling step, by specifying the interaction between the decision rule and the class-specific effect, and by preserving the source identity throughout the process. These improvements deal with the issues of inference and diagnostics in the specified context. The locked release-evaluation approach broadens the positive stress ranking comparison to new speakers and to a new set of attacks.

However, the ranking differences for the held-out portion are lower than those for the development portion, but the pooling effect on the reduction in FAKE miss is higher. The different populations and denominators make it impossible to estimate a controlled effect of change in partitioning. Also, the development errors in A05 and A06 cannot be extrapolated to A07 through A19. There is information in the official documentation of the corpus that links the different attack algorithms used in different partitions, and the five-column metadata does not establish the complete lineage of the generators. Therefore, the scientific conclusion should be made based on the designated partitions and conditions.

8 Limitations and Threats to Validity

Nested training sets are balanced and compute-constrained. Training on full-foundation data is out of scope for the current experiment design and would alter the class composition to be 89.83% FAKE rather than 50% FAKE. The comparison of sizes makes use of only one initialization and one nested expansion that doubles updates from 840 to 1,680. Questions about size effects at

fixed compute and robustness under different draws of the cohort remain open. Both A and B have limited speaker resources; development attack sets contain 18 or 19 sources.

The three development initialization blocks use the identical development data and checkpointing rule. The identical partition is used for checkpointing selection and development analysis reporting. The results are dependent on the predetermined decision threshold and checkpoint used. Specifically, the guard failure refers to the selected checkpoint comparison. It does not mean that all checkpoints or epochs would fail.

The matched architecture design involved retraining just a single architecture. Preprocessing, noise rescaling, and codec reaction can independently modify score difference without affecting forensic cue preservation. At present, no independently validated forensic target exists for use in our experiments. The locked within-release evaluation used four fits of one architecture trained on 640 recordings each. Some attacks share algorithms with earlier partitions. An external corpus with known audio labels, source lineage, and distinct speakers/utterances must be available to test cross-corpus generalization.

8.1 Internal and external validity

The signal intervention manages the designed processing and the matched inputs, yet the clean population decides which speakers and synthesis instances are permitted to participate. In the case of release, reuse of attack families makes it impossible for there to be any simple unseen-generator interpretation. Additionally, the detector picks checkpoints based on the clean development BCE, which may combine the clean performance with the selection data. The selection registration precludes any retrospective modification of the stress effects, without altering the dependence. Other operating points require an independent locked design.

The source is the unit used to link ordered views, while the speaker is the resampling unit. The model receives correlated transformations of the same speech signal at evaluation time. More views do not substitute for more speakers. Specifically, attack-level confidence intervals depend on a small number of clusters of sources and speakers. Another specification of the bootstrap, another sample, or another leave-one-speaker-out approach could affect the size of the interval.

8.2 Engineering scope

Reproducibility is guaranteed by execution integrity, deterministic replay verification, and artifact sealing in the case of the fitted comparisons. They do not guarantee that there is an equivalent number or latency behavior in the production version. These results have been measured using a CPU running in the registered environment and including training and validation operations. These results do not prove anything about mobile latency, energy consumption, streaming, or adversarial adaptation. The complete evaluation is based on a bounded and balanced set of attacks for each release.

8.3 Construct validity and paired uncertainty

A FAKE score is an output of the classifier. It will be affected by transformation if the latter changes the spectral structure, the temporal texture, the signal scale or the range of the

preprocessing input. In the absence of an independent definition of the cue and intervention, the difference in scores cannot differentiate between the transformation of evidence and of sensitivity to that evidence.

The estimates from within-seed pairing correspond to a test population that is shared and rely on the registered clusters of speakers. The chosen clean checkpoint is an additional element of variation since different realizations of the training set could lead to choosing a different epoch. Differences between three blocks serve as robustness checks but cannot provide any reliable estimation of distribution of the results with respect to seed, architecture, and/or corpus. Thus, for each development contrast separately, mean and sample standard deviation are reported.

The held-out test sample consists of 64 sources for each FAKE attack and 256 REAL sources from 67 speakers. The choice of quota sampling differs from the official partition in terms of the weight of attacks. The attack intervals are pointwise and do not address multiplicity correction. Selection of A17-A19 and breakdown of particular mistakes was an afterthought and therefore any analysis of subgroups is exploratory, even when particular intervals exclude zero. These sources were initially rated with the fixed design. The choice of a model on these scores will erase the distinction.

9 Conclusion

In RQ1, channel augmentation boosted the stressed ranking in all the three 640-development blocks and the original locked within-release comparisons. The relatively smaller ranking gain at 1,280 in seed 9207 accompanied fewer stressed FAKE misses and more stressed REAL false alarms at threshold 0.5. The clean-BCE-selected Channel checkpoint violated both clean-risk guards.

For RQ2, the A05/A06 miss comparisons inverted between the two size protocols in seed 9207 while the initializations varied in 640. The original held out pairs still suffered from clean FAKE misses and stressed misses on A17-A19. For RQ3, inversion of noise and codec order affected the decision for the same recordings and the REAL order contrast depended on the training protocol used. Reporting ranking along with classwise risks, attack counts, and source-specific transitions reveals such differences. In the size comparison, one seed is used together with more optimizer steps and shared development sources. Transfer between corpora and mechanism behind the response remains to be explained.

Reproducibility

The experiment registries, per-source scores, checkpoint seals, and speaker-bootstrap outputs document eight development fits and the original four-model locked evaluation. The supplement identifies the four selected seed-9207 checkpoints used for the size contrast and provides the registered endpoints, pointwise speaker intervals, order transitions and execution costs. The analysis code reproduces the size contrast from the sealed development scores. It also records the three-seed 640 analysis and the retrospective threshold and standalone component diagnostics. Section 5 defines the evaluation populations.

Acknowledgement

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